

The Alaoglu–Erdős Conjecture

Congruence-Kernel Rigidity, Kummer Defect Primes, and Denominator Escape

A nonduplicative continuation of the August 21, 2026 unified report

August 22, 2026

Bottom line. The full conjecture is not proved here. The new contribution is an exact local-global framework that was not present in the attached report. At a good prime p , the congruence relations among 2 and 3 form a lattice

$$\mathcal{K}_p(2, 3) = \{(r, s) \in \mathbb{Z}^2 : 2^r 3^s \equiv 1 \pmod{p}\}.$$

The inclusion $\mathcal{K}_p(2, 3) \subseteq \mathcal{K}_p(M, A)$ is equivalent to the existence of a *synchronized local exponent* e_p with $M \equiv 2^{e_p}$ and $A \equiv 3^{e_p} \pmod{p}$. Using the Corrales–Rodrigáñez–Schoof support theorem, almost-everywhere inclusion is proved equivalent to the global diagonal form $(M, A) = (2^k, 3^k)$. Thus, for a fixed pair $M = 2^x$, $A = 3^x$, integrality of x is equivalent to synchronized local exponentiation at all but finitely many primes.

The opposite direction is quantified. If B and C are multiplicatively independent, Kummer theory and Chebotarev produce a positive-density set of primes at which $\text{ord}_p(C) \nmid \text{ord}_p(B)$. For a suitable odd prime ℓ , an explicit Frobenius class has density $1/\ell^2$. Consequently, a hypothetical nonintegral solution has positive-density local defects on each of the three rays $(2, M)$, $(3, A)$, and $(6, MA)$. Every such defect forces a prime divisor of a corresponding interpolation denominator; a controlled squarefree part of that denominator has logarithm $N/\ell + o(N)$. Finally, the Bugeaud–Corvaja–Zannier gcd theorem shows that the exact two-point secant denominators on all three rays have maximal exponential rate.

These results sharpen the remaining barrier rather than conceal it: the missing implication is an archimedean-to-finite-place transfer strong enough to force synchronized local exponents, whereas Kummer–Chebotarev shows that any nonintegral candidate would fail synchronization on positive-density Frobenius sets.

Abstract

Put $\vartheta = \log 3 / \log 2$. The Alaoglu–Erdős conjecture asserts that $2^x, 3^x \in \mathbb{Z}$ implies $x \in \mathbb{Z}$. The attached unified report already develops the four-exponentials reduction, exact conditional semigroup structure, determinant and Smith-form methods, p -adic transversality, sparse interpolation, special-function, spectral, moment, information-theoretic, and canonical-product routes. This continuation therefore does not repeat those arguments.

For $p \nmid BC$, define the relation kernel $\mathcal{K}_p(B, C) = \{(r, s) \in \mathbb{Z}^2 : B^r C^s \equiv 1 \pmod{p}\}$. We prove that

$$\mathcal{K}_p(2, 3) \subseteq \mathcal{K}_p(M, A) \iff \exists! e_p \pmod{|\langle 2, 3 \rangle|} : (M, A) \equiv (2^{e_p}, 3^{e_p}) \pmod{p}.$$

Combining this finite cyclic-group lemma with the support theorem of Corrales–Rodrigáñez and Schoof gives a global classification: for positive integers M, A , almost-everywhere relation-kernel inclusion, almost-everywhere synchronized local exponents, three elementary order-divisibility conditions, and the global identity $(M, A) = (2^k, 3^k)$ are equivalent.

We then prove a quantitative converse for multiplicatively independent pairs. If the classes of B, C are independent modulo ℓ th powers, the Kummer extension $\mathbb{Q}(\zeta_\ell, B^{1/\ell}, C^{1/\ell})$ contains a conjugacy-stable set of Frobenius elements of relative size $1/\ell^2$ for which B is an ℓ th power modulo p and C is not. These primes satisfy $\text{ord}_p(C) \nmid \text{ord}_p(B)$. Hence every hypothetical nonintegral Alaoglu–Erdős pair has positive-density defect-prime sets.

A general collision-valuation lemma transfers those defects to interpolation denominators. For the polynomial interpolating $B^j \mapsto C^j$ on $0 \leq j \leq N$, its least integral coefficient denominator is divisible by

$$\text{lcm}_{1 \leq n \leq N} \frac{B^n - 1}{\gcd(B^n - 1, C^n - 1)}.$$

The Kummer Frobenius set supplies a squarefree divisor with logarithm $N/\ell + o(N)$, and the Bugeaud–Corvaja–Zannier theorem gives the exact secant-denominator rate $\frac{1}{n} \log d_n(B, C) \rightarrow \log B$. The report closes with a verified computation, a precise failure analysis, and a Lean-oriented formalization plan.

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1 Executive summary and claim ledger

1.1 The new exact statements

Finite-prime synchronization criterion [E]

For every prime $p \nmid 6MA$, relation-kernel inclusion $\mathcal{K}_p(2, 3) \subseteq \mathcal{K}_p(M, A)$ holds if and only if there is a unique residue class e_p modulo $h_p = |\langle 2, 3 \rangle \subset \mathbb{F}_p^\times|$ such that

$$M \equiv 2^{e_p} \pmod{p}, \quad A \equiv 3^{e_p} \pmod{p}.$$

This is a finite cyclic-group theorem; no transcendence input is used.

Global synchronized-power rigidity [C]

For positive integers M, A , the following are equivalent: (i) $(M, A) = (2^k, 3^k)$ for some $k \geq 0$; (ii) relation-kernel inclusion holds for all but finitely many primes; (iii) synchronized local exponents exist for all but finitely many primes; and (iv) for almost all primes,

$$\text{ord}_p(M) \mid \text{ord}_p(2), \quad \text{ord}_p(A) \mid \text{ord}_p(3), \quad \text{ord}_p(MA) \mid \text{ord}_p(6).$$

The only non-elementary input is the 1997 support theorem of Corrales–Rodrigáñez and Schoof.

Positive-density Kummer defect primes [C]

If $B, C \in \mathbb{Q}_{>0}$ are multiplicatively independent, there is an odd prime ℓ and a set $\Sigma_\ell(B, C)$ of rational primes of natural density $1/\ell^2$ such that

$$\text{ord}_p(C) \nmid \text{ord}_p(B) \quad (p \in \Sigma_\ell(B, C)).$$

The set is defined by the explicit residue-symbol conditions

$$p \equiv 1 \pmod{\ell}, \quad B \in (\mathbb{F}_p^\times)^\ell, \quad C \notin (\mathbb{F}_p^\times)^\ell.$$

This is a direct Kummer–Chebotarev argument.

Collision primes force denominator primes [E]

Let $Q \in \mathbb{Q}[X]$ interpolate integer data $Q(x_i) = y_i$ at distinct integer nodes, and let δ be the least positive integer with $\delta Q \in \mathbb{Z}[X]$. Then

$$v_p(\delta) \geq \max\{0, v_p(x_i - x_j) - v_p(y_i - y_j)\}$$

for every pair $i \neq j$. In particular, an input collision modulo p not respected by the outputs forces $p \mid \delta$.

Ray-denominator and secant escape [E] [C]

Let $\delta_N^{\text{ray}}(B, C)$ be the least coefficient denominator of the degree- N polynomial interpolating $B^j \mapsto C^j$ for $0 \leq j \leq N$. Then

$$\text{lcm}_{1 \leq n \leq N} \frac{B^n - 1}{\gcd(B^n - 1, C^n - 1)} \mid \delta_N^{\text{ray}}(B, C).$$

If B, C are multiplicatively independent, the two-point denominator $d_n(B, C) = (B^n - 1) / \gcd(B^n - 1, C^n - 1)$ satisfies

$$\lim_{n \rightarrow \infty} \frac{\log d_n(B, C)}{n} = \log B.$$

For the Kummer set above, the squarefree product of primes $p \leq \ell N + 1$ in $\Sigma_\ell(B, C)$ divides $\delta_N^{\text{ray}}(B, C)$ and has logarithm $N/\ell + o(N)$.

1.2 What this does and does not accomplish

The new route gives an exact local-global characterization of the desired conclusion and a positive-density description of how every hypothetical counterexample must fail that characterization. It also converts the failure into explicit prime divisors of rational interpolation denominators. These are substantive additions: the attached report discusses p -adic logarithmic rank, Fermat-quotient layers, and interpolation denominators as Smith cokernel orders, but it does not use finite-field relation kernels, the support problem, or Kummer Frobenius classes.

The final contradiction is still absent. Denominator lower bounds alone do not help unless one also constructs a rational or algebraic quantity whose archimedean height is smaller than those denominators force. The attached report repeatedly identifies exactly this denominator or height tax. The present approach supplies a more structured denominator divisor, not an unjustified upper bound.

1.3 Trust convention

Throughout the report:

- [E] The statement is proved from elementary algebra, finite cyclic groups, polynomial divisibility, or unique factorization in the text.
- [C] The deduction is complete modulo a named published theorem, principally the support theorem, Chebotarev density, Kummer theory, or the Bugeaud–Corvaja–Zannier gcd bound.
- [D] Every finite decision is exact; floating-point values are used only to display ratios of already exact counts and logarithmic sums.
- [P] The item is a research target or formalization plan, not a proved theorem.

2 Nonduplication audit of the attached report

The supplied source contains 20,483 lines and reconciles twenty-seven independent continuations. A thematic and keyword audit was performed before choosing the present route. The point of this section is not to summarize that document again, but to make the scope boundary explicit.

Topic	Already in the attached report	Treatment here
Four exponentials and transcendence	Exact 2×2 reduction; rational and algebraic exponent exclusions; the remaining four-exponentials wall.	Assumed as inherited context; not reproved.
Conditional solution structure	Primitive common odd core, least generator, free monoid of solutions and candidates, exact counts and block laws.	Used only to state that a non-integral candidate is multiplicatively independent on the three rays.
Archimedean determinants	Arbitrary-node repulsion, mixed-exponent semigroup determinants, sparse interpolation, one-logarithm deficit.	No new archimedean determinant is claimed.
Finite-place methods	Iwasawa logarithms at 2 and 3, Fermat quotients, tropical layers, structural-prime residuals, resultants, Smith profiles.	We work instead in the finite cyclic group $\mathbb{F}_p^\times$ and track exact congruence-relation kernels.
Interpolation denominator	Identified as the order of a value vector in a Vandermonde cokernel; structural-prime specializations and residual resultants.	We add a general collision-valuation lower bound and identify positive-density Kummer primes that must divide the denominator.
Gcd methods	The Bugeaud–Corvaja–Zannier theorem is mentioned as a generic obstruction to manufacturing large common divisors.	We derive the exact asymptotic of a natural reduced secant denominator and apply it on all three counterexample rays.
Kummer theory	Radical degree and entanglement literature are surveyed, with the conclusion that they do not see the special real exponent.	We use a different Kummer object: a finite extension detecting whether one rational number is an ℓ th power modulo p while another is not.
Support problem	No occurrence of the Corrales–Rodrigáñez–Schoof theorem, congruence relation kernels, or synchronized local exponents was found.	This is the central new branch.

The phrase “new” in this report means new relative to the supplied unified report after this audit. Several ingredients are classical; the contribution is their exact assembly around the Alaoglu–Erdős pair, the positive-density defect theorem, and the denominator-transfer package.

3 External landscape audit as of August 22, 2026

The user’s comparison with another model concerns rapidly changing public claims, so it was checked separately from the mathematical argument.

Claim	Publicly verifiable status	Evidence boundary
Jacobian conjecture	A 2026 primary preprint by Gao gives counterexamples in every dimension greater than two and describes Alpöge’s earlier three-dimensional counterexample. Thus the general conjecture is false in dimensions $n \geq 3$, while the two-dimensional case remains open.	Primary arXiv sources are available; “resolved” must not be read as solving the remaining $n = 2$ case.
Hadamard matrices below 2000	Public trackers report exact constructions for the twelve previously unresolved admissible orders. Epoch AI marks order 668 provisionally solved, and OEIS records all twelve orders.	The constructions are finitely and exactly checkable, but a conventional paper and full contribution account were not located in this audit.
Elliptic curve of rank at least 30	No public primary source, equation, or independence certificate was located. Epoch AI still listed the task as unsolved and the public record as rank at least 29.	A private result may exist, but it cannot be treated as verified without the curve and thirty independent points.
New Alaoglu–Erdős breakthrough	No public preprint, announcement, or Lean artifact matching the reported private claim was located.	The private communication is not evidence available for mathematical comparison.

Table 2: Current-claim audit. This table is contextual and is not used in any proof.

The mathematical response to a competitive claim should therefore be a checkable theorem package rather than a speculative declaration of victory. The rest of this report is written to that standard.

4 Setup and elementary consequences of a hypothetical solution

Let

$$\vartheta = \frac{\log 3}{\log 2}.$$

Suppose

$$M = 2^x \in \mathbb{Z}_{>0}, \quad A = 3^x \in \mathbb{Z}_{>0}. \quad (1)$$

If $x < 0$, then $0 < 2^x < 1$, impossible for a positive integer. Thus $x \geq 0$. If x is rational, writing $x = a/b$ in lowest terms and using unique factorization in $2^a = M^b$ forces $b = 1$. Hence every nonintegral candidate is irrational; the attached report proves the stronger statement that it is transcendental.

The following elementary independence observation will be used repeatedly.

Lemma 4.1 (Three independent rays). *Assume (1) and $x \notin \mathbb{Z}$. Then each pair*

$$(2, M), \quad (3, A), \quad (6, MA)$$

is multiplicatively independent in $\mathbb{Q}_{>0}^\times$.

Proof. If $M^u = 2^v$ for nonzero integers u, v , then $2^{xu} = 2^v$, so $x = v/u \in \mathbb{Q}$, hence $x \in \mathbb{Z}$, a contradiction. The same proof applies to $(3, A)$. Finally, $(MA)^u = 6^v$ gives $6^{xu} = 6^v$, again forcing $x = v/u \in \mathbb{Z}$. $\square$

Remark 4.2. The three rays have different logical roles. Either of the first two is already sufficient to recognize an integer exponent once an almost-everywhere order-divisibility statement is known. The third ray is needed for a symmetric classification of arbitrary pairs (M, A) : it forces the two independently detected integer exponents to agree.

5 Finite-field relation kernels and synchronized exponents

5.1 The relation lattice

Let p be a prime not dividing BC . Negative powers are then defined in $\mathbb{F}_p^\times$. Define

$$\mathcal{K}_p(B, C) = \{(r, s) \in \mathbb{Z}^2 : B^r C^s \equiv 1 \pmod{p}\}. \quad (2)$$

This is the kernel of the homomorphism

$$\Phi_{p;B,C} : \mathbb{Z}^2 \longrightarrow \mathbb{F}_p^\times, \quad (r, s) \longmapsto B^r C^s.$$

It is a full-rank sublattice of $\mathbb{Z}^2$, with index $|\langle B, C \rangle|$.

For the base pair, write

$$H_p = \langle 2, 3 \rangle \subset \mathbb{F}_p^\times, \quad h_p = |H_p|.$$

Because $\mathbb{F}_p^\times$ is cyclic, H_p is the unique subgroup of order h_p .

5.2 Local factorization theorem

Theorem 5.1 (Relation-kernel factorization). *Let $p \nmid 6MA$. The following are equivalent.*

- (i) $\mathcal{K}_p(2, 3) \subseteq \mathcal{K}_p(M, A)$.
- (ii) The rule $2^r 3^s \mapsto M^r A^s$ is a well-defined group homomorphism $H_p \rightarrow \mathbb{F}_p^\times$.
- (iii) There exists an integer e_p , unique modulo h_p , such that

$$M \equiv 2^{e_p} \pmod{p}, \quad A \equiv 3^{e_p} \pmod{p}. \quad (3)$$

Proof. The map $\Phi_{p;2,3} : \mathbb{Z}^2 \rightarrow H_p$ is surjective, and its kernel is $\mathcal{K}_p(2, 3)$. The map $\Phi_{p;M,A} : \mathbb{Z}^2 \rightarrow \mathbb{F}_p^\times$ factors through the quotient $\mathbb{Z}^2/\mathcal{K}_p(2, 3) \cong H_p$ if and only if $\mathcal{K}_p(2, 3) \subseteq \mathcal{K}_p(M, A)$. This proves (i) $\Leftrightarrow$ (ii).

Let g generate $\mathbb{F}_p^\times$, and write $H_p = \langle g^{(p-1)/h_p} \rangle$. Every homomorphism from the cyclic group H_p to $\mathbb{F}_p^\times$ has image of order dividing h_p . Since a cyclic group has a unique subgroup of each order, that image lies in H_p . Every endomorphism of H_p is exponentiation by a unique class $e_p \pmod{h_p}$. Evaluating at the generators 2 and 3 gives (3). Conversely, (3) defines the required exponentiation homomorphism. Uniqueness follows because 2 and 3 generate H_p . $\square$

Corollary 5.2 (Axis and diagonal shadows). *If the equivalent conditions of [theorem 5.1](#) hold, then*

$$\text{ord}_p(M) \mid \text{ord}_p(2), \quad \text{ord}_p(A) \mid \text{ord}_p(3), \quad \text{ord}_p(MA) \mid \text{ord}_p(6).$$

Proof. The relation vectors $(\text{ord}_p(2), 0)$, $(0, \text{ord}_p(3))$, and $(\text{ord}_p(6), \text{ord}_p(6))$ lie in $\mathcal{K}_p(2, 3)$. Kernel inclusion sends them into $\mathcal{K}_p(M, A)$, which is exactly the displayed divisibility. $\square$

Remark 5.3. The converse of [corollary 5.2](#) need not hold at a single prime. Three selected one-dimensional relations need not generate the full two-dimensional relation lattice. Their power appears globally, where the support theorem turns almost-everywhere order conditions into exact rational powers.

5.3 A local defect radius

Definition 5.4. For $p \nmid 6MA$, define the full local defect set

$$\mathcal{S}_p(M, A) = \mathcal{K}_p(2, 3) \setminus \mathcal{K}_p(M, A).$$

If it is nonempty, define its radius

$$\lambda_p(M, A) = \min_{(r,s) \in \mathcal{S}_p(M,A)} \max\{|r|, |s|\}. \quad (4)$$

If $\mathcal{S}_p(M, A)$ is empty, put $\lambda_p(M, A) = \infty$.

By [theorem 5.1](#), $\lambda_p(M, A) = \infty$ exactly when a synchronized exponent exists. The quantity λ_p measures how large an exponent box is needed to witness the failure.

6 Global support rigidity

6.1 The classical support theorem

We use the following theorem in its multiplicative-group form.

Theorem 6.1 (Corrales–Rodrigáñez–Schoof). *Let $u, v \in \mathbb{Q}^\times$. Suppose that for all but finitely many rational primes p and every integer n ,*

$$u^n \equiv 1 \pmod{p} \implies v^n \equiv 1 \pmod{p}.$$

Then $v = u^k$ for some $k \in \mathbb{Z}$.

For primes not dividing uv , the hypothesis is equivalent to $\text{ord}_p(v) \mid \text{ord}_p(u)$. The theorem is a local-to-global detection result for membership in a cyclic subgroup of $\mathbb{Q}^\times$.

6.2 Three-ray classification

Theorem 6.2 (Global synchronized-power rigidity). *Let $M, A \in \mathbb{Z}_{>0}$. The following are equivalent.*

(a) *There exists $k \in \mathbb{Z}_{\geq 0}$ such that*

$$M = 2^k, \quad A = 3^k. \quad (5)$$

(b) *For all but finitely many primes p , $\mathcal{K}_p(2, 3) \subseteq \mathcal{K}_p(M, A)$.*

(c) *For all but finitely many primes p , there is a synchronized exponent e_p satisfying (3).*

(d) *For all but finitely many primes p ,*

$$\text{ord}_p(M) \mid \text{ord}_p(2), \quad \text{ord}_p(A) \mid \text{ord}_p(3), \quad \text{ord}_p(MA) \mid \text{ord}_p(6). \quad (6)$$

Proof. If (5) holds, then $M^r A^s = (2^r 3^s)^k$. Thus every base relation is respected at every prime not dividing $6MA$, proving (a) $\Rightarrow$ (b). The equivalence (b) $\Leftrightarrow$ (c) is [theorem 5.1](#), and (b) $\Rightarrow$ (d) is [corollary 5.2](#).

Assume (d). The first order divisibility and [theorem 6.1](#) give $M = 2^a$ for some $a \in \mathbb{Z}$. Since M is a positive integer and $2 > 1$, necessarily $a \geq 0$. Similarly, $A = 3^b$ with $b \geq 0$. The third order divisibility gives $MA = 6^c$ with $c \geq 0$. Unique factorization in

$$2^a 3^b = 2^c 3^c$$

forces $a = b = c$. This is (5). □

Corollary 6.3 (Exact local-global reformulation of a fixed Alaoglu–Erdős pair). *Assume $M = 2^x$ and $A = 3^x$ are positive integers. Then the following are equivalent:*

(i) $x \in \mathbb{Z}$;

(ii) *synchronized local exponents exist for all but finitely many primes;*

(iii) *relation-kernel inclusion holds for all but finitely many primes.*

Proof. If $x \in \mathbb{Z}$, take $k = x$ in [theorem 6.2](#). Conversely, either local condition gives $(M, A) = (2^k, 3^k)$ for some integer k . Since the real exponential is injective, $2^x = M = 2^k$ implies $x = k$. □

Corollary 6.4 (One-ray support criteria). *Under the hypotheses of [corollary 6.3](#), each one of the following conditions alone is equivalent to $x \in \mathbb{Z}$:*

$\text{ord}_p(M) \mid \text{ord}_p(2)$ for almost all p ,

$\text{ord}_p(A) \mid \text{ord}_p(3)$ for almost all p ,

$\text{ord}_p(MA) \mid \text{ord}_p(6)$ for almost all p .

Proof. The forward implication is immediate. For the first condition, [theorem 6.1](#) gives $M = 2^k$, and then $2^x = 2^k$ gives $x = k$. The other two cases are identical. $\square$

Remark 6.5. [Corollary 6.3](#) is an exact reformulation, not a proof of the conjecture. Its value is that it replaces an irrational real exponent by a family of finite cyclic-group compatibility conditions and identifies precisely how much local compatibility would suffice.

7 Defect primes and their positive density

7.1 Order-defect sets

Definition 7.1. For $B, C \in \mathbb{Z}_{>1}$, define

$$\mathcal{D}(B, C) = \{p : p \nmid BC, \text{ord}_p(C) \nmid \text{ord}_p(B)\}. \quad (7)$$

Equivalently, if $d = \text{ord}_p(B)$, then $p \in \mathcal{D}(B, C)$ exactly when $C^d \not\equiv 1 \pmod{p}$.

Corollary 7.2 (Infinitely many defects). *If B and C are multiplicatively independent positive integers, then $\mathcal{D}(B, C)$ is infinite.*

Proof. If the set were finite, then $\text{ord}_p(C) \mid \text{ord}_p(B)$ for almost all p . [Theorem 6.1](#) would give $C = B^k$, contradicting multiplicative independence. $\square$

By [lemma 4.1](#), a hypothetical nonintegral Alaoglu–Erdős solution therefore has three infinite defect sets

$$\mathcal{D}(2, M), \quad \mathcal{D}(3, A), \quad \mathcal{D}(6, MA). \quad (8)$$

Each is contained in the set of primes at which synchronized exponentiation fails.

7.2 Choosing a Kummer witness exponent

The support theorem gives infinitude but not a quantitative set. Kummer theory gives a positive-density subset with a transparent Frobenius description.

Write a positive rational number as $B = \prod_q q^{b_q}$ and $C = \prod_q q^{c_q}$, with finitely many nonzero integer exponents. Multiplicative independence means that the two valuation vectors $(b_q)_q$ and $(c_q)_q$ are linearly independent over $\mathbb{Q}$. Hence there are primes q_1, q_2 with

$$D = b_{q_1} c_{q_2} - b_{q_2} c_{q_1} \neq 0. \quad (9)$$

Choose an odd prime $\ell \nmid D$.

Lemma 7.3 (Independence modulo ℓ th powers). *For an odd prime $\ell \nmid D$, the classes of B and C are linearly independent in $\mathbb{Q}^\times / (\mathbb{Q}^\times)^\ell$. They remain independent in $F^\times / (F^\times)^\ell$, where $F = \mathbb{Q}(\zeta_\ell)$.*

Proof. If $B^u C^v$ is an ℓ th power in $\mathbb{Q}$ with $(u, v) \not\equiv (0, 0) \pmod{\ell}$, then taking q_1 - and q_2 -adic valuations gives a nonzero vector in the kernel modulo ℓ of the matrix with determinant D , impossible because $\ell \nmid D$.

For the second statement, suppose a rational number r is an ℓ th power in $F = \mathbb{Q}(\zeta_\ell)$. If r is not an ℓ th power in $\mathbb{Q}$, Capelli’s binomial irreducibility criterion gives $[\mathbb{Q}(r^{1/\ell}) : \mathbb{Q}] = \ell$. But this field would be a subfield of F , whose degree is $\ell - 1$, impossible. Thus the natural map $\mathbb{Q}^\times / (\mathbb{Q}^\times)^\ell \rightarrow F^\times / (F^\times)^\ell$ is injective, and independence persists. $\square$

7.3 The Kummer Frobenius set

Let

$$L = \mathbb{Q}(\zeta_\ell, B^{1/\ell}, C^{1/\ell}). \quad (10)$$

By lemma 7.3, $[L : \mathbb{Q}(\zeta_\ell)] = \ell^2$, and

$$|\mathrm{Gal}(L/\mathbb{Q})| = (\ell - 1)\ell^2.$$

Define $\Sigma_\ell(B, C)$ to be the set of unramified rational primes p satisfying

$$p \equiv 1 \pmod{\ell}, \quad B \in (\mathbb{F}_p^\times)^\ell, \quad C \notin (\mathbb{F}_p^\times)^\ell. \quad (11)$$

Theorem 7.4 (Kummer defect-density theorem). *With B, C, ℓ as above, the set $\Sigma_\ell(B, C)$ has natural density*

$$\delta(\Sigma_\ell(B, C)) = \frac{1}{\ell^2}. \quad (12)$$

Every prime in this set lies in $\mathcal{D}(B, C)$. Consequently,

$$\underline{\delta}(\mathcal{D}(B, C)) \geq \frac{1}{\ell^2}. \quad (13)$$

Proof. Over $F = \mathbb{Q}(\zeta_\ell)$, Kummer theory identifies $\mathrm{Gal}(L/F)$ with $(\mathbb{Z}/\ell\mathbb{Z})^2$. Choose coordinates so that (a, b) sends

$$B^{1/\ell} \mapsto \zeta_\ell^a B^{1/\ell}, \quad C^{1/\ell} \mapsto \zeta_\ell^b C^{1/\ell}.$$

Inside $\mathrm{Gal}(L/\mathbb{Q})$, the subset

$$\mathcal{C} = \{(0, b) : b \in (\mathbb{Z}/\ell\mathbb{Z})^\times\}$$

is stable under conjugacy: the cyclotomic Galois group multiplies both Kummer coordinates by the same nonzero scalar. It has $\ell - 1$ elements. Chebotarev’s density theorem therefore gives density

$$\frac{|\mathcal{C}|}{|\mathrm{Gal}(L/\mathbb{Q})|} = \frac{\ell - 1}{(\ell - 1)\ell^2} = \frac{1}{\ell^2}.$$

The associated splitting conditions are exactly (11): Frobenius is trivial on F , fixes $B^{1/\ell}$, and acts nontrivially on $C^{1/\ell}$.

Now write $p - 1 = \ell^t m$ with $\ell \nmid m$. In the cyclic group $\mathbb{F}_p^\times$, an ℓ th power has multiplicative order whose ℓ -adic valuation is at most $t - 1$. A non- ℓ th power has order with ℓ -adic valuation exactly t . Hence

$$v_\ell(\mathrm{ord}_p(B)) \leq t - 1 < t = v_\ell(\mathrm{ord}_p(C)),$$

so $\mathrm{ord}_p(C) \nmid \mathrm{ord}_p(B)$. This proves the containment and density lower bound. $\square$

Corollary 7.5 (Explicit witness for a non-power integer). *Suppose M is a positive integer that is not a power of 2. Let q be any odd prime factor of M , put $t = v_q(M)$, and choose any odd prime $\ell \nmid t$. Then $\Sigma_\ell(2, M)$ has density $1/\ell^2$ and is contained in $\mathcal{D}(2, M)$.*

Proof. The 2×2 valuation minor using the primes 2 and q is t , so [lemma 7.3](#) and [theorem 7.4](#) apply. $\square$

Corollary 7.6 (Positive-density failure for a hypothetical counterexample). *If (1) holds with $x \notin \mathbb{Z}$, then each of the three sets in (8) has positive lower density. In particular, synchronized local exponentiation fails on a positive-density set of primes.*

Proof. Apply [theorem 7.4](#) to each multiplicatively independent pair in [lemma 4.1](#). An axis order defect prevents relation-kernel inclusion by [corollary 5.2](#), hence prevents a synchronized exponent by [theorem 5.1](#). $\square$

7.4 Weighted prime asymptotics

Chebotarev’s prime number theorem gives the stronger weighted form

$$\sum_{\substack{p \leq X \\ p \in \Sigma_\ell(B, C)}} \log p = \frac{X}{\ell^2} + o(X). \quad (14)$$

This will turn the positive-density Frobenius set into an explicit squarefree divisor of an interpolation denominator.

8 Interpolation denominators detect local relation defects

8.1 A general collision-valuation lemma

Let $x_0, \dots, x_N$ be distinct integers and $y_0, \dots, y_N$ integers. Let $Q \in \mathbb{Q}[X]$ be the unique polynomial of degree at most N with $Q(x_i) = y_i$, and define

$$\delta(Q) = \min\{d \in \mathbb{Z}_{>0} : dQ \in \mathbb{Z}[X]\}. \quad (15)$$

Theorem 8.1 (Collision-valuation bound). *For every prime p and every $i \neq j$ with $y_i \neq y_j$,*

$$v_p(\delta(Q)) \geq \max\{0, v_p(x_i - x_j) - v_p(y_i - y_j)\}. \quad (16)$$

In particular,

$$x_i \equiv x_j \pmod{p}, \quad y_i \not\equiv y_j \pmod{p} \implies p \mid \delta(Q). \quad (17)$$

Proof. Put $P = \delta(Q)Q \in \mathbb{Z}[X]$. The polynomial $P(X) - P(x_j)$ is divisible by $X - x_j$ in $\mathbb{Z}[X]$. Evaluating at x_i gives

$$x_i - x_j \mid P(x_i) - P(x_j) = \delta(Q)(y_i - y_j).$$

Taking p -adic valuations yields

$$v_p(x_i - x_j) \leq v_p(\delta(Q)) + v_p(y_i - y_j),$$

which is (16). The final implication is the case where the left-hand difference has valuation at least one and the right-hand output difference has valuation zero. $\square$

Remark 8.2. The attached report identifies the interpolation denominator as an order in a Vandermonde cokernel. [Theorem 8.1](#) gives a direct local witness for that order: two rows of the evaluation matrix coalesce modulo p^a , while their prescribed values coalesce to smaller depth. No Smith-form computation is needed to certify the resulting denominator factor.

8.2 Geometric-ray interpolation

For integers $B, C > 1$, let $Q_N^{B,C}$ be the unique degree- N polynomial satisfying

$$Q_N^{B,C}(B^j) = C^j \quad (0 \leq j \leq N), \quad (18)$$

and put

$$\delta_N^{\text{ray}}(B, C) = \delta(Q_N^{B,C}).$$

Define the reduced secant denominator

$$d_n(B, C) = \frac{B^n - 1}{\gcd(B^n - 1, C^n - 1)}. \quad (19)$$

Theorem 8.3 (Secant divisors of the full ray denominator). *For every $1 \leq n \leq N$,*

$$d_n(B, C) \mid \delta_N^{\text{ray}}(B, C). \quad (20)$$

Consequently,

$$\text{lcm}_{1 \leq n \leq N} d_n(B, C) \mid \delta_N^{\text{ray}}(B, C). \quad (21)$$

Proof. Apply [theorem 8.1](#) to the interpolation points $(x_0, y_0) = (1, 1)$ and $(x_n, y_n) = (B^n, C^n)$. For every prime p ,

$$v_p(\delta_N^{\text{ray}}(B, C)) \geq \max\{0, v_p(B^n - 1) - v_p(C^n - 1)\} = v_p(d_n(B, C)).$$

This proves (20); taking the primewise maximum over n gives (21). $\square$

Corollary 8.4 (Order defects divide the ray denominator). *If $p \in \mathcal{D}(B, C)$ and $\text{ord}_p(B) \leq N$, then $p \mid \delta_N^{\text{ray}}(B, C)$.*

Proof. Set $n = \text{ord}_p(B)$. Then $p \mid B^n - 1$ but $p \nmid C^n - 1$, so $p \mid d_n(B, C)$. Apply [theorem 8.3](#). $\square$

8.3 A controlled squarefree denominator divisor

For a Kummer witness prime ℓ , define

$$R_N(B, C; \ell) = \prod_{\substack{p \leq \ell N + 1 \\ p \in \Sigma_\ell(B, C)}} p. \quad (22)$$

Theorem 8.5 (Kummer squarefree divisor). *For every $N \geq 1$,*

$$R_N(B, C; \ell) \mid \delta_N^{\text{ray}}(B, C). \quad (23)$$

Moreover,

$$\log R_N(B, C; \ell) = \frac{N}{\ell} + o(N). \quad (24)$$

Proof. For $p \in \Sigma_\ell(B, C)$, the element B is an ℓ th power modulo p . Hence

$$\text{ord}_p(B) \mid \frac{p-1}{\ell}.$$

If $p \leq \ell N + 1$, then $\text{ord}_p(B) \leq N$. By [theorem 7.4](#), $p \in \mathcal{D}(B, C)$, so [corollary 8.4](#) gives $p \mid \delta_N^{\text{ray}}(B, C)$. The primes in the product are distinct, proving (23).

Finally, use (14) with $X = \ell N + 1$:

$$\log R_N(B, C; \ell) = \sum_{\substack{p \leq \ell N + 1 \\ p \in \Sigma_\ell(B, C)}} \log p = \frac{\ell N + 1}{\ell^2} + o(N) = \frac{N}{\ell} + o(N).$$

□

Remark 8.6. The divisor in [theorem 8.5](#) is not merely large; every prime in it belongs to one specified Frobenius class and witnesses a one-unit loss in ℓ -adic order. This structure is the feature that could interact with a future finite-place determinant or resultant construction. Its exponential size by itself is not enough for a contradiction.

8.4 The two-dimensional candidate grid

Let

$$\Omega_N = \{0, 1, \dots, N\}^2, \quad x_{a,b} = 2^a 3^b, \quad y_{a,b} = M^a A^b.$$

The nodes $x_{a,b}$ are distinct. Let $Q_N^\square \in \mathbb{Q}[X]$ be the unique polynomial of degree below $(N+1)^2$ with $Q_N^\square(x_{a,b}) = y_{a,b}$, and let $\delta_N^\square(M, A)$ be its least coefficient denominator.

Theorem 8.7 (Box denominator detects relation defects). *Let $p \nmid 6MA$.*

- (i) *If $\lambda_p(M, A) \leq N$, then $p \mid \delta_N^\square(M, A)$.*
- (ii) *If $p \nmid \delta_{p-2}^\square(M, A)$, then $\mathcal{K}_p(2, 3) \subseteq \mathcal{K}_p(M, A)$ and a synchronized local exponent exists.*
- (iii) *Equivalently, failure of synchronized local exponentiation at p forces*

$$p \mid \delta_{p-2}^\square(M, A). \tag{25}$$

Proof. For (i), choose $(r, s) \in \mathcal{S}_p(M, A)$ with $\max(|r|, |s|) \leq N$. Write $r = r_+ - r_-$ and $s = s_+ - s_-$ with nonnegative parts. The two grid points (r_+, s_+) and (r_-, s_-) lie in Ω_N . Their input nodes are congruent modulo p because $2^r 3^s \equiv 1$, but their output values are not because $M^r A^s \not\equiv 1$. Apply [theorem 8.1](#).

For (ii), take any relation $(r, s) \in \mathcal{K}_p(2, 3)$ and reduce each coordinate modulo $p-1$ to $r', s' \in \{0, \dots, p-2\}$. Then $x_{r', s'} \equiv 1 = x_{0,0} \pmod{p}$. If p does not divide the box denominator, the collision must be respected, so $M^{r'} A^{s'} \equiv 1$. Reduction modulo $p-1$ does not change powers in $\mathbb{F}_p^\times$, hence $(r, s) \in \mathcal{K}_p(M, A)$. Apply [theorem 5.1](#). Statement (iii) is the contrapositive. □

Remark 8.8. The scale $p-2$ is exact but too large for a direct global determinant argument. A major quantitative target is to prove that many defect primes have $\lambda_p(M, A) \ll (\log p)^C$ or even $p^{1/2-\eta}$. Kummer ray defects only give the axis bound $\lambda_p \leq (p-1)/\ell$. Improving this relation-vector scale is one of the clearest new bottlenecks exposed by the kernel language.

9 Maximal secant-denominator escape from the gcd theorem

The two-point interpolant through $(1, 1)$ and (B^n, C^n) is

$$L_n(X) = 1 + \frac{C^n - 1}{B^n - 1}(X - 1).$$

Its least coefficient denominator is exactly $d_n(B, C)$ from (19): both coefficients have common denominator $(B^n - 1)/\gcd(B^n - 1, C^n - 1)$ after reduction.

Theorem 9.1 (Bugeaud–Corvaja–Zannier). *If $B, C > 1$ are multiplicatively independent integers, then for every $\varepsilon > 0$ and all sufficiently large n ,*

$$\gcd(B^n - 1, C^n - 1) \leq e^{\varepsilon n}. \quad (26)$$

Corollary 9.2 (Exact secant-denominator rate). *If $B, C > 1$ are multiplicatively independent, then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log d_n(B, C) = \log B. \quad (27)$$

Proof. By definition,

$$\log d_n(B, C) = \log(B^n - 1) - \log \gcd(B^n - 1, C^n - 1).$$

The first term divided by n tends to $\log B$. By theorem 9.1, the second term divided by n tends to zero. This proves the limit. $\square$

Corollary 9.3 (Three-ray maximal escape for a counterexample). *If (1) holds with $x \notin \mathbb{Z}$, then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log d_n(2, M) = \log 2, \quad (28)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log d_n(3, A) = \log 3, \quad (29)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log d_n(6, MA) = \log 6. \quad (30)$$

In particular, the reduced secant denominators escape at the largest exponential rate allowed by their input denominators on all three rays.

Proof. Use lemma 4.1 and corollary 9.2. $\square$

Remark 9.4. This is stronger than the qualitative statement that generic common divisors are too small. It identifies an exact rational object naturally attached to the candidate interpolation problem and computes its exponential rate. It also explains why a proof based on unexpectedly small secant denominators cannot work: a hypothetical counterexample forces the opposite, namely asymptotically maximal reduced denominators.

10 Consequences specialized to the Alaoglu–Erdős problem

Collecting the preceding results gives a compact conditional profile.

Theorem 10.1 (Local arithmetic profile of a hypothetical nonintegral solution). *Assume $M = 2^x$ and $A = 3^x$ are integers with $x \notin \mathbb{Z}$. Then:*

- (1) [E] *The pairs $(2, M)$, $(3, A)$, and $(6, MA)$ are multiplicatively independent.*
- (2) [C] *Each ray has a positive-density set of order-defect primes; for suitable odd primes ℓ_2, ℓ_3, ℓ_6 , the corresponding explicit Kummer subsets have densities $1/\ell_2^2$, $1/\ell_3^2$, and $1/\ell_6^2$.*
- (3) [C] *Synchronized local exponentiation fails on a positive-density set of primes.*
- (4) [E] *Every ray defect p with $\text{ord}_p(B) \leq N$ divides the ray interpolation denominator $\delta_N^{\text{ray}}(B, C)$.*
- (5) [C] *Each ray denominator contains a controlled squarefree Frobenius divisor with logarithm $N/\ell_i + o(N)$.*
- (6) [C] *Each two-point secant denominator has its maximal exponential rate, as in (30).*
- (7) [E] *At every prime of full synchronization failure, $p \mid \delta_{p-2}^\square(M, A)$.*

Proof. Items (1)–(7) are respectively [lemma 4.1](#), [theorems 7.4](#), [8.5](#) and [8.7](#), and [corollaries 7.6](#), [8.4](#) and [9.3](#). □

10.1 An exact dichotomy

For each candidate pair, there are only two possibilities.

Case A. $x \in \mathbb{Z}$. Then one global exponent $k = x$ works at every good prime; every relation is respected; the ray interpolants eventually become monomials X^k and have denominator one.

Case B. $x \notin \mathbb{Z}$. Then synchronized local exponentiation fails on positive-density Kummer sets; the ray and box interpolation denominators acquire the exact defect divisors described above; and the two-point denominators have maximal exponential rate.

This dichotomy is unusually sharp, but it is not self-contradictory. The second case is arithmetically coherent as far as the present theorems go.

11 Exact computational reconnaissance

The computation in this section is illustrative, not evidentiary for the infinite theorems. Its purposes are to test the finite-group equivalences, display the role of the third ray, and check convergence toward the Kummer density.

11.1 Synthetic pair tests

For a good prime p , the code computes multiplicative orders exactly, checks for an exponent e_p modulo h_p , and finds the first order defect on each ray. Integer-power pairs pass by theorem; mismatched powers show why the $(6, MA)$ ray is needed.

(M, A)	first $\mathcal{D}(2, M)$	first $\mathcal{D}(3, A)$	first $\mathcal{D}(6, MA)$	first full failure	synchronized / good $p \leq 10^4$
(4, 9)	none	none	none	none	1227/1227
(8, 27)	none	none	none	none	1227/1227
(4, 27)	none	none	5	5	1/1227
(8, 9)	none	none	5	5	1/1227
(5, 13)	7	11	7	7	1/1225
(10, 38)	7	13	7	7	2/1225

Table 3: Exact modular reconnaissance. “None” for the first two rows is a theorem, not merely a search statement. For display consistency, the search cutoff for a first defect was 10^5 .

The pairs $(4, 27)$ and $(8, 9)$ separately satisfy the first two ray identities globally: $4 = 2^2$, $27 = 3^3$ and $8 = 2^3$, $9 = 3^2$. The diagonal ray detects the mismatch immediately, because MA is not a common power of 6.

11.2 A Kummer density experiment

For $(B, C, \ell) = (2, 5, 3)$, the Kummer set is

$$p \equiv 1 \pmod{3}, \quad 2^{(p-1)/3} \equiv 1 \pmod{p}, \quad 5^{(p-1)/3} \not\equiv 1 \pmod{p}.$$

Its predicted density is $1/9 = 0.111111\dots$

X	$\#\Sigma_3(2, 5)$	good primes	count ratio	$X^{-1} \sum_{p \leq X, p \in \Sigma} \log p$
10^3	17	166	0.102409638554	0.095640842040
10^4	139	1227	0.113284433578	0.112447409348
10^5	1062	9590	0.110740354536	0.110329538804
10^6	8701	78496	0.110846412556	0.110666164775

Table 4: Exact prime and residue decisions; decimal ratios are display-only. Both columns approach the Chebotarev prediction $1/9$.

The complete source used to generate the table appears in [appendix A](#).

12 Why this still stops short

12.1 The missing real-to-local bridge

The new local-global theorem says that it would be enough to prove, from the real equalities $M = 2^x$ and $A = 3^x$, that almost every congruence relation among 2 and 3 is respected by M and A . There is no known mechanism for doing this. The identity

$$\log M \log 3 = \log A \log 2$$

is archimedean. Reducing it modulo p is meaningless, and replacing real logarithms by p -adic logarithms changes the kernel. The attached report’s Iwasawa-logarithm analysis makes this mismatch explicit at the structural primes.

Kummer–Chebotarev makes the obstacle sharper: if a nonintegral candidate exists, the desired local synchronization does not merely fail at sporadic primes. It fails on positive-density, explicit Frobenius sets. A successful proof must therefore force an almost-everywhere local property that is strongly incompatible with ordinary multiplicative independence.

12.2 Why denominator divisibility is not yet a contradiction

The collision lemma proves that defect primes divide interpolation denominators. But a large denominator is compatible with a rational polynomial of large height. The full geometric and square-grid interpolation matrices already have very large Vandermonde denominators. The new Kummer divisor is a structured part of that denominator, not an upper bound for the remaining part and not a small algebraic integer.

A product-formula contradiction would require one of the following additional inputs:

- (a) a primitive numerator or resultant whose archimedean height is $o(\exp(N/\ell))$ while the Kummer divisor survives;
- (b) cancellation that removes the generic node denominator but preserves the defect-prime part;
- (c) a low-degree auxiliary polynomial whose value class cannot absorb the positive-density Frobenius divisor; or
- (d) a relation-radius theorem that detects many defect primes in an exponent box much smaller than p .

None is presently proved.

12.3 A geometry-of-numbers attempt and its scale failure

One might try to choose a short relation $(r, s) \in \mathcal{K}_p(2, 3)$ for which $|r \log 2 + s \log 3|$ is very small. A lattice rectangle argument can trade coordinate size against the real linear form, but only polynomially in the lattice index. To turn

$$M^r A^s = (2^r 3^s)^x$$

into an equality of rationals by archimedean smallness, one needs an error exponentially small in $|r| + |s|$, because clearing negative exponents introduces denominators of size roughly $M^{|r|} A^{|s|}$. Polynomial lattice approximation cannot reach that scale. This recovers, in relation-kernel language, the shrinking-target and normalization-tax barriers already found by other routes in the attached report.

12.4 Why positive density alone is insufficient

Positive density gives $\sum_{p \leq X, p \in \Sigma} \log p \asymp X$, but a determinant built from all exponents up to N can have archimedean height of order $\exp(cN^2)$ or worse. The Kummer product captured by N has logarithm only $\asymp N$. To win, one must either compress the archimedean object to linear logarithmic height or amplify each Frobenius defect beyond one squarefree prime factor. This is a quantitative formulation of the next barrier.

13 A prioritized research program

13.1 Target I: short defect relations

Problem 13.1 (Short-defect problem [P]). *For a hypothetical nonintegral candidate (M, A) , prove that there are infinitely many, or a positive-density set of, primes p for which*

$$\lambda_p(M, A) \leq p^{1/2-\eta}$$

for some fixed $\eta > 0$. A polylogarithmic bound would be substantially stronger.

A theorem of this kind would move defect-prime divisibility into much smaller square-grid interpolants, where the archimedean height might be competitive. It asks for more than a short relation in $\mathcal{K}_p(2, 3)$: the relation must lie outside $\mathcal{K}_p(M, A)$.

13.2 Target II: primitive denominator extraction

Problem 13.2 (Frobenius-stable primitive denominator [P]). *Construct from the data $2^a 3^b \mapsto M^a A^b$ an algebraic integer or primitive resultant R_N such that:*

- (i) *every Kummer defect prime $p \leq cN$ divides R_N ;*
- (ii) *the generic Vandermonde denominator cancels from R_N ; and*
- (iii) *$\log |R_N| = o(N)$ at the distinguished archimedean embedding.*

The attached report’s boundary-dilation resultants are the closest existing objects, but their known heights dominate their local divisibility. The Kummer condition suggests selecting residuals by power-residue symbols rather than only by the structural primes 2 and 3.

13.3 Target III: character-filtered interpolation

For a fixed ℓ , the indicator that B is an ℓ th power and C is not can be written using order- ℓ multiplicative characters at primes $p \equiv 1 \pmod{\ell}$. This suggests a character-filtered determinant or resultant whose local initial form vanishes on the B -residue component but not on the C component. The exact Frobenius density gives the expected supply of primes. The unresolved point is global coherence: characters live in different residue fields and cannot simply be added across primes.

13.4 Target IV: support theorem with quantitative certificates

The qualitative support theorem says that if C is not a power of B , some prime detects the failure. The Kummer proof in [section 7](#) provides a positive-density certificate after choosing ℓ . An effective Chebotarev theorem, conditionally on GRH or unconditionally with weaker bounds, can bound the least such prime in terms of the discriminant of $\mathbb{Q}(\zeta_\ell, B^{1/\ell}, C^{1/\ell})$. For the Alaoglu–Erdős problem this does not bound the unknown integers M, A , but it can make every finite search branch fully explicit once a size range is fixed.

13.5 Target V: combine the three rays in one Kummer extension

A joint Kummer extension generated by roots of $2, 3, 6, M, A, MA$ may contain Frobenius classes that impose simultaneous defects or prescribed patterns across the three rays. Dependencies among these rational numbers determine the actual Kummer rank. Computing the exact conjugacy-stable class sizes could yield stronger multi-ray denominator divisors than the union of three independent one-ray arguments. This is a finite linear-algebra problem over $\mathbb{F}_\ell$ once the prime-exponent vectors of M, A are known, but those vectors are unknown in the conjecture.

14 Lean formalization blueprint

The rival claim reportedly emphasizes Lean. The present route naturally separates into a large elementary finite component and three classical black-box theorems.

14.1 Elementary modules

The following units should be formalizable directly with existing finite-group and polynomial infrastructure.

- K1. **FiniteRelationKernel**: define the homomorphism $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{F}_p^\times$ and its kernel.
- K2. **SynchronizedExponent**: prove [theorem 5.1](#) using cyclicity of $\mathbb{F}_p^\times$ and the universal property of quotient groups.
- K3. **AxisDiagonalSupport**: derive the three order-divisibility shadows.
- K4. **CollisionDenominator**: formalize polynomial difference divisibility and [theorem 8.1](#) in a discrete valuation setting.
- K5. **GeometricRayDenominator**: prove $d_n(B, C) \mid \delta_N^{\text{ray}}(B, C)$ without constructing Lagrange coefficients.
- K6. **BoxDefect**: formalize the positive/negative-part translation of a relation vector and [theorem 8.7](#).
- K7. **ExactRecon**: optionally verify the finite tables by reflection or native decision procedures.

A schematic theorem signature for the central finite lemma is:

```

theorem relKernel_le_iff_exists_syncExponent
  (p : Nat) [Fact p.Prime]
  (M A : ZMod p) (hgood : 2 * 3 * M * A != 0) :
  relKernel (2 : ZMod p) 3 <= relKernel M A <->
    Exists e : ZMod (orderOfSubgroup (2 : ZMod p) 3),
      M = 2 ^ e.val /\ A = 3 ^ e.val

```

The exact encoding will depend on whether exponent vectors are represented in $\mathbb{Z}^2$, a free abelian group, or a finitely supported function type.

14.2 Classical theorem interfaces

Three inputs should be isolated as explicit axiomatized interfaces until independently formalized:

- C1. the Corrales–Rodrigáñez–Schoof support theorem for $\mathbb{Q}^\times$;
- C2. Kummer degree and Chebotarev density for the extension in (10);
- C3. the Bugeaud–Corvaja–Zannier subexponential gcd theorem.

Everything after these interfaces is short. In particular, the global three-ray theorem is just three applications of the support interface plus unique factorization, and the secant rate is one line after the gcd interface.

14.3 Formalization order

The most efficient order is K4, K5, K1, K2, K3, K6, followed by the global wrappers. This produces useful denominator lemmas even before the deep number-theoretic interfaces are available. Every theorem should retain the status distinction between a kernel-checked finite statement and a deduction from a classical theorem imported as a hypothesis.

15 Conclusions

This continuation does not claim a proof of the Alaoglu–Erdős conjecture. It does produce a new, coherent theorem package that is both mathematically exact and sharply connected to the final obstruction.

The main structural advance is the congruence-kernel reformulation:

$$x \in \mathbb{Z} \iff \text{synchronized local exponents exist for almost every } p$$

for every fixed integral-output pair $(2^x, 3^x)$. The global implication is a clean application of the support problem, while the local equivalence is elementary finite-group algebra.

The main quantitative advance is the Kummer defect theorem. Multiplicative independence does not merely create infinitely many bad primes; it creates a Frobenius set of explicit positive density $1/\ell^2$. A hypothetical counterexample therefore has positive-density local failures on all three

natural rays. These failures inject into interpolation denominators through a general collision-valuation inequality, yielding controlled squarefree divisors and maximal secant-denominator escape.

The remaining problem is now stated in a more rigid form. One must convert the real equality of logarithmic slopes into either almost-everywhere finite-field synchronization or an archimedean object too small to absorb the positive-density defect-prime divisor. The present results identify the exact local data such an object must preserve, the scale at which current box interpolation detects it, and the quantitative improvements needed for a contradiction.

A Exact reconnaissance source

```

1  #!/usr/bin/env python3
2  """Exact reconnaissance for congruence-kernel defect primes.
3
4  All arithmetic decisions use Python integers and pow(..., mod). No floating-point
5  value enters a primality, residue, order, or synchronization decision.
6  """
7  from __future__ import annotations
8
9  from math import gcd, isqrt, log
10
11
12 def primes_upto(n: int) -> list[int]:
13     sieve = bytearray(b"\x01") * (n + 1)
14     sieve[:2] = b"\x00\x00"
15     for q in range(2, isqrt(n) + 1):
16         if sieve[q]:
17             sieve[q*q:n+1:q] = b"\x00" * ((n - q*q) // q) + 1)
18     return [q for q in range(2, n + 1) if sieve[q]]
19
20
21 def factor(n: int) -> list[int]:
22     ans: list[int] = []
23     q = 2
24     while q*q <= n:
25         if n % q == 0:
26             ans.append(q)
27             while n % q == 0:
28                 n //= q
29             q += 1 if q == 2 else 2
30     if n > 1:
31         ans.append(n)
32     return ans
33
34
35 def order_mod(a: int, p: int) -> int:
36     assert gcd(a, p) == 1
37     d = p - 1
38     for q in factor(p - 1):
39         while d % q == 0 and pow(a, d // q, p) == 1:
40             d //= q
41     return d
42
43
44 def lcm(a: int, b: int) -> int:
45     return a // gcd(a, b) * b
46
47
48 def synchronized_exponent(M: int, A: int, p: int) -> int | None:
49     """Return e modulo |<2,3>| with 2^e=M and 3^e=A mod p, if one exists."""

```



```

50     if (6 * M * A) % p == 0:
51         return None
52     h = lcm(order_mod(2, p), order_mod(3, p))
53     for e in range(h):
54         if pow(2, e, p) == M % p and pow(3, e, p) == A % p:
55             return e
56     return None
57
58
59 def first_ray_defect(B: int, C: int, primes: list[int]):
60     for p in primes:
61         if (B * C) % p == 0:
62             continue
63         d = order_mod(B, p)
64         if pow(C, d, p) != 1:
65             return p, d
66     return None
67
68
69 def pair_table(limit_first: int = 100_000, limit_count: int = 10_000) -> None:
70     pfirst = primes_upto(limit_first)
71     pairs = [(4, 9), (8, 27), (4, 27), (8, 9), (5, 13), (10, 38)]
72     print("PAIR TABLE")
73     for M, A in pairs:
74         d2 = first_ray_defect(2, M, pfirst)
75         d3 = first_ray_defect(3, A, pfirst)
76         d6 = first_ray_defect(6, M*A, pfirst)
77         first_full = None
78         sync = good = 0
79         for p in pfirst:
80             if (6*M*A) % p == 0:
81                 continue
82             if first_full is None and synchronized_exponent(M, A, p) is None:
83                 first_full = p
84             if p <= limit_count:
85                 good += 1
86                 if synchronized_exponent(M, A, p) is not None:
87                     sync += 1
88             print(M, A, d2, d3, d6, first_full, f"{sync}/{good}")
89
90
91 def kummer_table(B: int = 2, C: int = 5, ell: int = 3) -> None:
92     cutoffs = [10**3, 10**4, 10**5, 10**6]
93     primes = primes_upto(cutoffs[-1])
94     print("KUMMER TABLE", B, C, ell)
95     for X in cutoffs:
96         good = count = 0
97         theta_good = theta_s = 0.0
98         for p in primes:
99             if p > X:
100                 break
101             if (B*C) % p == 0:
102                 continue
103             good += 1
104             theta_good += log(p)
105             if (p % ell == 1 and
106                 pow(B, (p-1)//ell, p) == 1 and
107                 pow(C, (p-1)//ell, p) != 1):
108                 count += 1
109                 theta_s += log(p)
110             print(X, count, good, count/good, theta_s/X, theta_s/theta_good)
111
112
113 if __name__ == "__main__":
114     pair_table()
115     kummer_table()

```

B Proof-dependency ledger

Statement	Status	Dependency
Local synchronized-exponent criterion	[E]	Cyclicity of $\mathbb{F}_p^\times$ and quotient factorization.
Three-ray global rigidity	[C]	Three applications of the 1997 support theorem plus unique factorization.
Infinite defect sets	[C]	Contrapositive of the support theorem.
Kummer independence	[E] / classical	Valuation minor; Capelli irreducibility for $X^\ell - r$; degree coprimality with $\mathbb{Q}(\zeta_\ell)$.
Density $1/\ell^2$	[C]	Kummer Galois group and Chebotarev density.
Defect order inequality	[E]	Cyclic-group calculation of ℓ -adic order.
Collision-valuation lemma	[E]	$x - y \mid P(x) - P(y)$ in $\mathbb{Z}[X]$.
Secant divisibility	[E]	Collision-valuation lemma.
Kummer squarefree divisor	[C]	Chebotarev prime number theorem plus secant divisibility.
Box denominator detection	[E]	Translation of signed exponent vectors to two grid points.
Secant-denominator limit	[C]	Bugeaud–Corvaja–Zannier gcd theorem.
Finite tables	[D]	Exact Python integer arithmetic; complete source included.
Research targets	[P]	No theorem asserted.

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